

Algebra 1: Unit 1, Lesson 6

1. Complete the statement to find the equivalence using laws of exponents.

$$\sqrt[4]{(6)^4} = ((6)^4)^{\frac{1}{4}} = (6)^{\frac{4}{4}} = \underline{\hspace{2cm}}$$

2. Using the first row as a guide, complete the table.

$(a)^{\frac{b}{c}}$	$(a^b)^{\frac{1}{c}}$	$\sqrt[c]{a^b}$
$\underline{\hspace{2cm}}$	$(17^4)^{\frac{1}{7}}$	$\sqrt[7]{17^4}$

3. The expression $(w)^{\frac{k}{g}}$ is in the form $(a)^{\frac{b}{c}}$. Rewrite the expression in the form $(a^b)^{\frac{1}{c}}$.